

A Physics-Based and Data-Driven Method for Aircraft Engine Remaining Useful Life Prediction

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Abstract

Remaining useful life (RUL) prediction of aircraft turbofan engines is an important problem with various real-world applications. Optimizing RUL prediction can improve safety and efficiency by allowing aircraft maintenance personnel to know the real-time health of the engine, which also increases cost-effectiveness. However, current models for RUL prediction have certain limitations, such as a lack of physical knowledge and overfitting to specific data. The two main types of models for RUL prediction currently, which are data-driven and physics-based, each suffers from their drawbacks. This paper proposes a combined physics-based and data-driven model in which a dataset with various aircraft engine sensor measurements is used in combination with a thermodynamic model in order to create a generalized RUL prediction model that minimizes some of the existing drawbacks. However, due to the unavailability of specific necessary data, this paper serves as a proof of concept (POC) for this model and outlines that this method is possible if the necessary data is available.

1 Introduction

The RUL of an aircraft engine is defined as the remaining amount of time that an engine can function properly until it fails and needs to be replaced [1]. This is an important metric in determining the health of an aircraft engine. RUL estimation is an important task in managing an engine's health, as it allows aircraft maintenance personnel to know the real-time health status of the engine and implement any necessary maintenance. This heavily reduces engine maintenance costs and avoids any unexpected problems, which is important for improving flight safety and efficiency.

There are multiple ways in which the RUL of an aircraft engine can be predicted using machine learning (ML). One of the most common methods is a data-driven approach [2]. In this method, past data related to the condition and components of the aircraft, as well as the current health state of the system is used to predict RUL. This is done by using large amounts of data such as sensor data from the aircraft engine to train an ML model, generally a neural network (NN), an ML model inspired by the structure and functioning of the human

brain, which is able to learn and make predictions through layers of interconnected nodes, known as neurons.

Another way to predict the RUL is through physics-informed methods, in which mechanistic knowledge specific to the system, condition monitoring data, and failure regulation are used to predict the RUL. However, there are several drawbacks to this method, including limited physical knowledge available to construct a model, and the difficulty to accurately determine parameters of the physical model. This makes data-driven methods generally the more popular option for predicting RUL.

Digital twins are defined as virtual models that simulate a certain physical piece of equipment [3]. This piece of equipment will typically have various sensors that provide information about the equipment’s performance, functionality, efficiency, health, and other factors. This data is applied to the virtual model to simulate the equipment’s performance over time and is constantly updated based on past and new data. While digital twins are similar to regular simulations, they are much more useful due to their ability to use historical and real-time data to provide more realistic representations of equipment and the environment. A digital twin works on a greater scale and with more detail compared to a regular simulation. This allows it to be an accurate real-time model as well as make predictions on the future of the equipment. There are a number of benefits regarding the use of digital twins, including more effective and efficient development of equipment, better testing and improvement of equipment, and improved maintenance and lifespan prediction of equipment.

One particularly important use case of digital twins is in the field of aerospace engineering. Companies can better engineer new parts and equipment by simulating how they perform in various conditions. They can also use digital twins to predict maintenance and failure of equipment and use data to extend lifespan.

NASA’s Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) is a tool for simulating large turbofan engine data of aircraft [4]. Through this tool, sensor data about an aircraft engine such as temperatures of various components and fan speeds can be extracted. This data can then be used to predict RUL through various methods and models.

This paper aims to serve as a POC for a method in which the C-MAPSS dataset is used to optimize the prediction of an aircraft turbofan engine’s RUL. The data provided in this dataset comes from a digital twin simulation of various aircraft engines and can be used to create ML models. As stated, the two most common ML methods to predict RUL are data-driven and physics-based methods. However, it is also possible to combine these two methods, which this paper will focus on. Due to existing drawbacks with both methods of RUL prediction, such as the need for large amounts of data and enough physical knowledge, a combination of data-driven models with physics-informed methods can help eliminate some of these drawbacks, which will be further explored in this paper. Specifically, this paper will address the drawback of data-driven models overfitting to the data they are trained on, and not being general enough to be used for a wider range of applications. However, due to limited access to certain necessary data, the full extent of this method cannot be explored. Hence, this paper establishes a method to predict RUL in which data-driven and physics-

based methods are combined. It does this by assuming values for data in order to establish a POC and affirms that if that data could be accessed, the method would be successful in optimizing and generalizing RUL prediction.

2 Background

This section outlines brief background information regarding aircraft turbofan engines and their components as well as applicable notation and equations relating to properties of turbofan engines.

2.1 Turbofan Engine

The turbofan engine is a modern variation of a jet engine in which thrust is produced using a combination of jet core efflux and bypass air that is accelerated by a ducted fan that is driven by the jet core [9]. The engine has six main components that work together to produce thrust. Below is a simple labeled diagram of a turbofan engine [10].

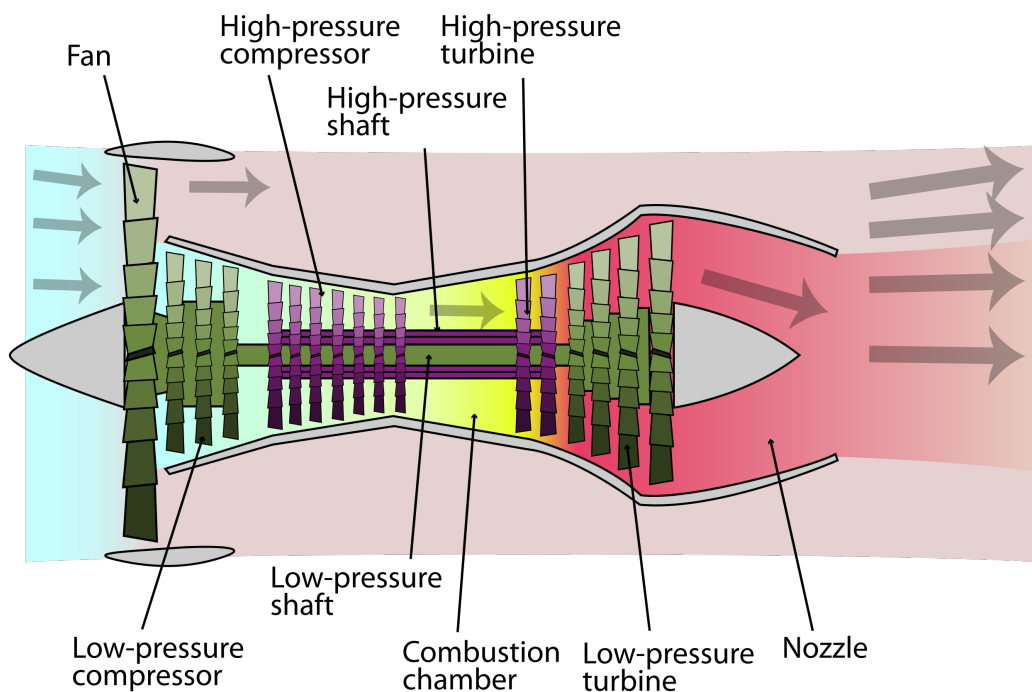


Figure 1: Simplified diagram of aircraft turbofan engine [10]

The engine works starting with the fan, whose purpose is to take in a large mass of air in the bypass flow and accelerate it [11]. This air is the main source of thrust production. In a geared turbofan engine, the fan uses a gearbox that allows the fan to be independent from other components such as the low-pressure compressor and turbine, which allows it to run at a specific speed separate from those other components, optimizing it for maximum

efficiency. The air then travels to a low-pressure compressor (LPC), which acts as a sort of pre-compressor. This means that it takes in air and compresses it in preparation for the high-pressure compressor (HPC). This is the main source of air compression, which is done right before the air is fed into the combustor. The compressed air flows into the combustor, where it is mixed with fuel and burned at a temperature close to 1700°C. This is the component that converts the air into a source of thrust.

Once the air is combusted, it flows into the turbine. This component takes the energy that is contained in the combusted gas and converts it into mechanical energy. The turbine is split into high-pressure (HPT) and low-pressure (LPT) components, with the low-pressure turbine generating the majority of the thrust. Finally, the gas flows out of the outlet nozzle.

2.2 Notation and Equations

In order to create the proposed model for RUL prediction, a number of equations need to be defined, which are used to calculate temperatures and pressures of various components of the engine, which is applicable to the physics component of the combined model. The notation used in these equations is outlined as follows [12]:

Symbol	Name
alt	Altitude
M_a	Mach number
f	Fuel flow
P_a	Ambient pressure
T_a	Ambient temperature
γ_c	Specific heat ratio of cold air
γ_h	Specific heat ratio of hot air
λ	Work done factor
η_d	Isentropic efficiency
η_f	Fan efficiency
π_f	Fan pressure ratio
η_{LPC}	LPC efficiency
π_{LPC}	LPC pressure ratio
η_{HPC}	HPC efficiency
π_{HPC}	HPC pressure ratio
η_{CC}	CC efficiency
C_{p_c}	Specific heat at bypass
C_{p_h}	Specific heat at core
Q_R	Caloric value of fuel
ΔP_{CC}	CC pressure loss

Table 1: Applicable notation for thermodynamic equations

There are six equations used in the physics-based component of the model. First, the

pressure and temperature at the inlet are calculated using the following equations:

$$P_{02} = P_a(1 + \eta_d \frac{\gamma_c - 1}{2} M_a^2)^{\frac{\gamma_c}{\gamma_c - 1}} \quad (1)$$

$$T_{02} = T_a(1 + \frac{\gamma_c - 1}{2} M_a^2) \quad (2)$$

Here, P_{02} is the inlet pressure, and T_{02} is the inlet temperature. γ_c is given a value of 1.4. Next, the bypass pressure is calculated with the following equation:

$$T_{010} = T_{02} \pi_f \quad (3)$$

Following that, the LPC and HPC temperatures are calculated with the following equations:

$$T_{03} = T_{02}(1 + \frac{1}{\pi_f}(\pi_f^{\frac{\gamma_c - 1}{\gamma_c}} - 1))(1 + \frac{1}{\eta_{LPC}}(\pi_{LPC}^{\frac{\gamma_h - 1}{\gamma_h}} - 1)) \quad (4)$$

$$T_{04} = T_{03}(1 + \frac{1}{\eta_{HPC}}(\pi_{HPC}^{\frac{\gamma_h - 1}{\gamma_h}} - 1)) \quad (5)$$

Here, T_{03} represents LPC temperature, T_{04} represents HPC temperature, and γ_h is given a value of 1.33. Next, the pressure at the HPT is calculated with the following equation:

$$P_{05} = P_{010} \pi_{LPC} \pi_{HPC} - \Delta P_{CC} \quad (6)$$

Finally, the temperature at the LPT is calculated with the following equation:

$$T_{06} = \frac{f \eta_{cc} Q_R + C_{pc} T_{04}}{C_{ph} + f C_{ph}} - \frac{C_{pc} (T_{04} - T_{03})}{\lambda C_{ph} (1 + f)} \quad (7)$$

3 Literature Review

There are a variety of papers that discuss specific methods of predicting the RUL of aircraft engines through various methods. One such paper, “A Framework of Dynamic Data Driven Digital Twin for Complex Engineering Products: the Example of Aircraft Engine Health Management”, proposes a data-driven model to predict RUL [5]. In this paper, a long short-term memory (LSTM) neural network was created. The model was trained and tested using data from various aircraft engine sensor measurements given by NASA’s C-MAPSS dataset. For this model specifically, measurements from 7 of the 21 available sensors were selected to be used in the model’s training and validation. The model’s performance was then evaluated using the metrics of root mean squared error (RMSE) and s-score and was shown to have performed better than both a similarity-based linear regression model and a feed-forward neural network (FFNN) in both metrics. In another paper, “Physics-based Remaining Useful Life Prediction for Aircraft Engine Bearing Prognosis”, a physics-based model is proposed to predict RUL, as opposed to the more common data-driven methods [6]. This method was developed in the Defense Advanced Research Projects Agency (DARPA) Engine System

Prognosis (ESP) system and faults in typical roller bearings, as spalls in roller bearings are one of the main causes of aircraft engine failure. When an unhealthy bearing is detected, a spall propagation model is used to predict the RUL, which is done by taking inputs of the spall size of the bearing and operating conditions to generate spall propagation trends. The RUL is then predicted by calculating when the trends pass a failure threshold and require maintenance.

These papers provide examples of data-driven and physics-based methods of predicting RUL, respectively. However, as stated, there are several drawbacks to each of these methods. Due to this, this paper proposes a model that includes a combination of data-driven and physics-based methods in an attempt to eliminate some drawbacks. In the paper “Fusing physics-based and deep learning models for prognostics”, a similar methodology is proposed [7]. In this paper, a model to predict RUL is created by combining a thermodynamic model with various machine learning models, and those combined models are compared with their respective purely data-driven models. The paper concluded that the combined models all performed better than the purely data-driven models. While the method proposed in this paper is similar, only certain data is extracted from C-MAPSS, and a thermodynamic model is more heavily relied on in order to mitigate over-reliance on certain data and to make the model more generally applicable.

In the paper “Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation”, the C-MAPSS data and its applications in predicting RUL are discussed [8]. The paper explains how the C-MAPSS data was acquired through the use of a digital-twin simulation of an aircraft engine, which took in 14 inputs, which were health parameters that allowed for the simulation of faults and deterioration in the engine’s various components. This produced 21 output variables, which were various sensor measurements including temperatures and pressures at various points in the engine. The paper then discusses the dataset’s applications in predicting RUL, as the dataset includes the sensor measurements for each cycle that the engines runs, as well as at what cycle the aircraft fails.

4 Methodology

This paper proposes a model built on a combination of physics-based and data-driven methods. This method is somewhat similar to the method used in [7] but differs in order to create a more general and widely applicable model. This is done by creating a dataset in which a set of sensor measurements and corresponding RUL is created for each engine in each dataset. The sensor measurements are calculated using a thermodynamic model rather than taken from the C-MAPSS data, allowing for a more generalized model. This data is then used to train and test a machine learning model, which can then be used to predict RUL.

4.1 Data

The first step to creating the model was collecting and processing the data. As stated, the data used for this method came from the C-MAPSS dataset. There were four training and testing datasets available. For each dataset, values of 21 different turbofan engine sensor measurements, provided by a digital twin simulation of a turbofan engine, along with the corresponding cycle value for that set of sensor measurements, were provided for a number of engines with differing efficiencies and pressure ratios. In addition, each dataset corresponded with a set of initial operating conditions, such as ambient pressure and temperature, which were parameters in the simulation that influenced the sensor measurements. The tables below outline the initial operating conditions and labels [14] and sensor measurements and labels [8] respectively.

Dataset	Alt (ft)	Mach	P_a (psia)	T_a ($^{\circ}F$)	f (pps)	Fan Speed (rpm)	Core Speed (rpm)
FC01	0	0	14.67	59	6.835	2388	9051
FC02	0	0.25	14.67	59	7.085	2403	9084
FC03	0	0.25	14.67	86	7.043	2432	9274
FC04	1000	0	14.18	59	6.567	2380	9021

Table 2: Initial operating conditions provided by C-MAPSS [14]

LABEL	NAME	UNIT
T2	Total temperature at fan inlet	$^{\circ}R$
T24	Total temperature at LPC outlet	$^{\circ}R$
T30	Total temperature at HPC outlet	$^{\circ}R$
T50	Total temperature at LPT outlet	$^{\circ}R$
P2	Pressure at fan inlet	psia
P15	Total pressure in bypass-duct	psia
P30	Total pressure at HPC outlet	psia
Nf	Physical fan speed	rpm
Nc	Physical core speed	rpm
epr	Engine pressure ratio (P50/P2)	–
Ps30	Static pressure at HPC outlet	psia
phi	Ratio of fuel flow to Ps30	pps/psi
NRf	Corrected fan speed	rpm
NRc	Corrected core speed	rpm
BPR	Bypass Ratio	–
farB	Burner fuel-air ratio	–
htBleed	Bleed Enthalpy	–
Nf_dmd	Demanded fan speed	rpm
PCNfR_dmd	Demanded corrected fan speed	rpm
W31	HPT coolant bleed	lbm/s
W32	LPT coolant bleed	lbm/s

Table 3: Sensor measurements with labels provided by C-MAPSS [8]

Because the C-MAPSS dataset provides the cycle at which an engine is for each row, and the data for one specific engine stops at some cycle value, the value at which the data for one engine stops and the next starts is that engine’s RUL, and the RUL at any cycle value can be calculated by subtracting the current cycle value from the total RUL. Below is an example plot of RUL values for each engine of the first training dataset (FD001).

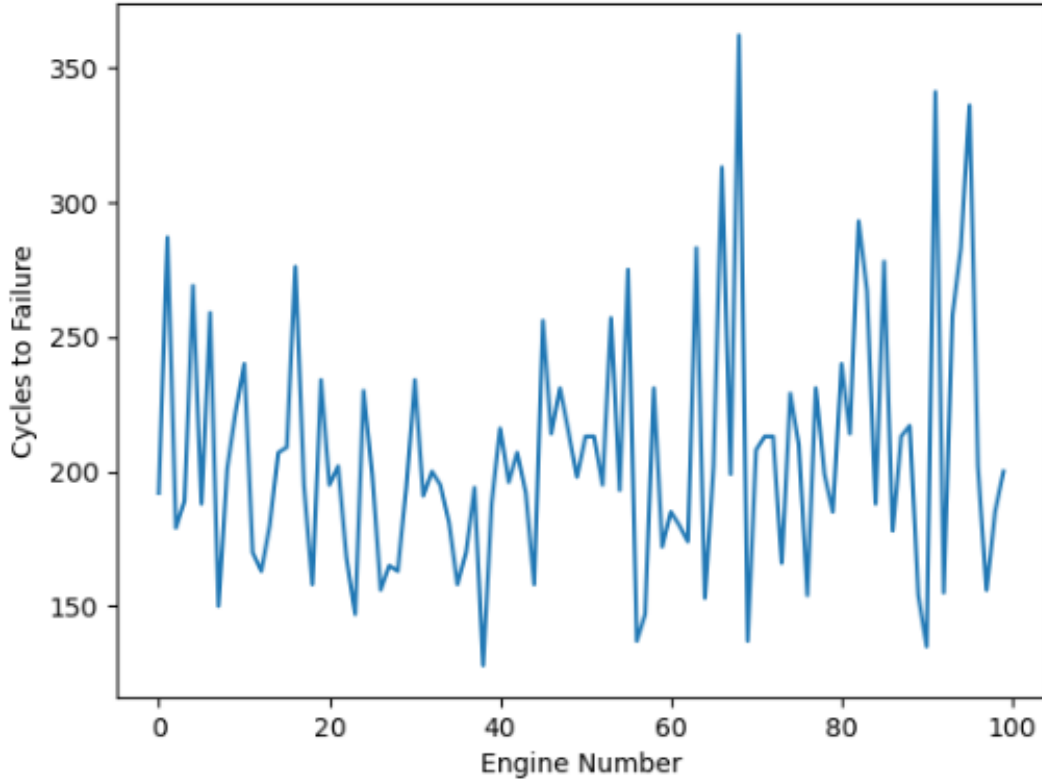


Figure 2: RUL vs engine for dataset 1

4.2 Thermodynamic Model

For this RUL prediction model, the sensor measurements from C-MAPSS are less important. Rather, the initial operating conditions, efficiency and pressure ratio values, and corresponding RULs are more important, as these are needed to create the thermodynamic model. However, a problem arises when attempting to obtain efficiency values. Because the C-MAPSS simulation itself is designed for government or institutional use and is not publicly available, the efficiency and pressure ratio values, which are inputs into the simulation, are not publicly available either. In order to address this problem, these values were assumed and randomly generated from a set of values and ranges provided in [12] and [13]. These values are listed in the table below.

Symbol	Value/Range
η_d	0.97 - 0.98
η_f	0.90 - 0.95
π_f	1.5482
η_{LPC}	0.85 - 0.91
π_{LPC}	1.49
η_{HPC}	0.85 - 0.91
π_{HPC}	20.7521
η_{CC}	0.9 - 0.98
C_{pc}	1005
C_{ph}	1148
Q_R	18500
ΔP_{CC}	0.02

Table 4: Assumed thermodynamic model parameter values/ranges

Because these values are assumptions and not the true values that correspond to the actual C-MAPSS data, the actual model created using these values will not be accurate, and will rather serve as a POC, stating that if these values were available, a working, accurate model could be created.

The thermodynamic model takes in the initial operating conditions, efficiency and pressure ratio values, and constants such as the specific heat ratio of cold air, and plugs those values into the six equations provided in section 2.2 to output a specific set of pressures and temperatures of various engine components. This set of outputs is decided based on the methodology in [7]. This is done for each engine in each dataset, set of outputs is paired with the corresponding RUL from C-MAPSS, in order to create a dataset that can be used in an ML model.

4.3 Machine Learning Model

Once the dataset is created, an ML model can be created to predict RUL. The ML model used in this method is an LSTM neural network, as it is a popular model used in aircraft engine RUL prediction, and is generally accurate with time-series data such as the data found in C-MAPSS. However, it is important to note that the specific type of neural network or ML model isn't as important, as this paper just demonstrates that this method of RUL prediction is possible. The basic architecture of the model is shown below [5]:

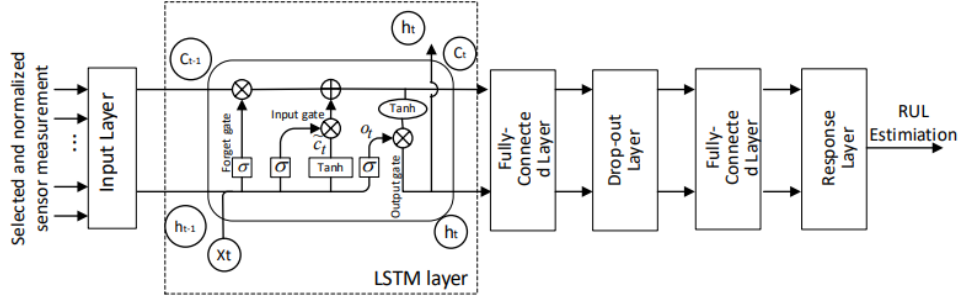


Figure 3: Architecture of LSTM neural network

Once the dataset is scaled using a standard scalar, it is inputted into the LSTM neural network with an 80%/20% train/test split. The neural network runs for 60 epochs, with a batch size of 20 and a dropout size of 0.2, meant to prevent overfitting.

5 Results

Below is a plot of the predicted and actual RUL for the test dataset, as well as the plot for the training and validation loss, where the blue line represents training loss and the orange line represents validation loss:

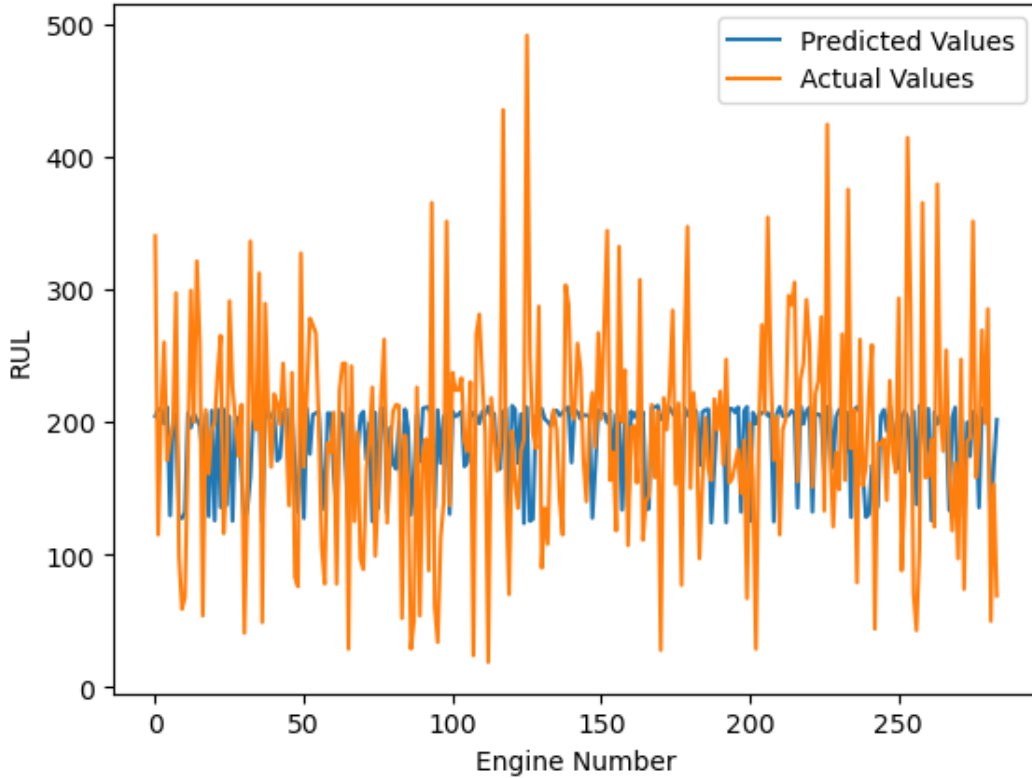


Figure 4: Predicted and actual RUL values for testing dataset

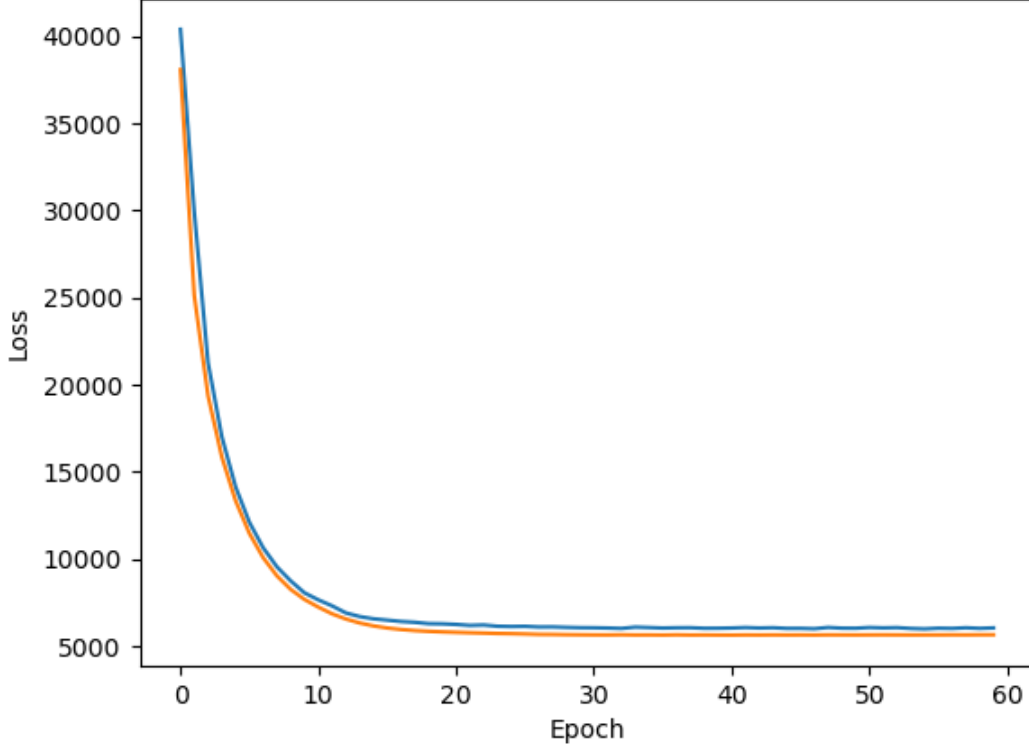


Figure 5: Model training and validation loss

A common metric used to evaluate model performance for RUL prediction, as well as neural networks in general, is RMSE. It can be defined in this context using the following equation [7]:

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (\widehat{RUL}_i - RUL_i)^2} \quad (8)$$

Here, m represents the total number of test data samples, $\widehat{RUL}_i$ represents the predicted RUL of sample i , and RUL_i represents the true RUL of sample i . The RMSE for this model was ~ 75.18 .

6 Discussion

As stated, it is necessary to note that because this is a POC with assumed parameter values, the true parameter values could not be determined. In addition, some of these parameter values were randomly generated between a specific range of values. This means that while the RUL differed between each engine with a general pattern, the outputted pressure and temperature values differed randomly between each original dataset. The only factors

that caused these values to change between each set of engines were the initial operating conditions, which also influenced the outputs.

With that noted, the model still managed to differ in predictions for each engine, as well as between each set of engines with differing initial operating conditions, as shown in Figure 4. In addition, over the training period of 60 epochs, both the training and validation decreased, as indicated in Figure 5. This shows that even with data that wasn't completely fit for training, the model still managed to train to some extent. This indicates that with accurate parameter values that accurately differ between engines, which would cause the outputted pressure and temperature values to differ as well, the model would train more accurately and have a much better performance. This is because the data used to train the model would correctly reflect the true state of each engine. In addition, the actual C-MAPSS data was not used to train the ML model. Instead, data based on concrete physics concepts and equations was created and used, meaning that this model, if real efficiency values were implemented, could be applied to much more general applications than a purely data-driven model could. This is because if only the data from C-MAPSS was used, the model would fit to that data and learn trends and patterns only found in that data. If that model was given new data without those same trends and patterns, it would not perform as well. Because physics equations and concepts were used to design this model, which are universal concepts, this model would perform the same with all types of data. As long as this model is given a set of initial operating conditions and engine efficiency and pressure ratio parameters, it would be able to accurately predict the engine's RUL.

The inaccurate assumed parameters also account for the relatively high RMSE value, which would also be improved if real values were implemented.

7 Conclusion

This paper proposed a proof of concept for a combined physics-based and data-driven model to improve the prediction of aircraft turbofan engine RUL. This is done by creating a generalized model that does not rely heavily on one set of data. Because of the limited availability of specific necessary data, the paper outlined a POC rather than a ready working model. The model was created by first acquiring data from NASA's C-MAPSS datasets. These provided a set of sensor measurements for a set of engines, their corresponding RUL values, and initial operating conditions, such as altitude and Mach number, that the C-MAPSS simulation was run using. Once the operating conditions and RUL for each engine were acquired, they were plugged into a thermodynamic model, along with a set of assumed efficiency and pressure ratio values given in [12] and [13]. This thermodynamic model, which consisted of six equations that calculated specific temperature and pressure values of specific components of a turbofan engine, outputted a dataset of a row of those values for each engine for each operating condition. This dataset, once scaled properly, was inputted into an LSTM neural network to be used as training and testing data. The trained ML model would not work as a complete model due to the fact that the efficiency and pressure ratio values were assumed and not accurate. However, the results, such as the training and validation loss, indicate

that if the missing values were available and used in the model, it would result in an adequate model. This model would be able to accurately predict the RUL of a turbofan engine given a set of initial operating conditions as well as efficiency and pressure ratio values.

If the necessary missing values could be accessed and implemented into the model, it would result in an accurate, working model. This could be used in real-world applications and applied more generally, and could predict when a real turbofan engine will fail. This would result in improved development, testing, and maintenance of aircraft turbofan engines.

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